

Jaiveer Bassi

707-656-6227 | jaiveerbassi@yahoo.com | [linkedin.com/in/jaiveer-bassi](https://www.linkedin.com/in/jaiveer-bassi) | github.com/imjbassi | imjbassi.github.io

Experience

Stanford University

Sep 2026 – Present

Undergraduate Research Fellow, Stanford Science Small Groups

Hybrid

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.
- Prepare research findings for a formal presentation to Stanford mentors and peers.

Physical Intelligence

May 2026 – Present

AI Robot Operator

San Francisco, CA

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.

Handshake AI

Apr 2026 – Jun 2026

Contracted AI Response Evaluator

Remote

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

Kaiser Permanente Regional Laboratory

Oct 2023 – Jun 2024

Software Developer

Berkeley, CA

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

Selected Projects

π 0 on a Budget | *OpenPI, LoRA, LeRobot, JAX, WSL2*

- Built an in-progress pipeline to record synchronized 30 fps camera frames and joint commands from a custom 4-DOF teleoperation arm and convert episodes into a LeRobot-compatible dataset.
- Implemented the consumer-GPU training, memory-profiling, open-loop evaluation, policy-serving, and closed-loop control workflow for adapting π 0-FAST on an RTX 4070. Real-data training awaits an upgrade from the current SG90 servos.

Fault-Tolerant Robot Telemetry Pipeline | *Kubernetes, Redis Streams, PostgreSQL*

- Designed a multi-stage telemetry pipeline for deliberate pod and network failures using durable acknowledgements, replay-safe writes, health probes, observability, and a chaos-testing harness.

CAN Bus Fault-Injection & Diagnostics Rig | *SocketCAN, MCP2515, Multimeter Diagnostics*

- Assembled a three-node CAN network with Arduino/MCP2515 nodes, a USB-CAN interface, and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

GELLO-Style Leader-Follower Teleoperation Rig | *Mechanical Assembly, Servo Control, Calibration*

- Built a four-DOF leader-follower mechanism for approximately \$30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

UR5e Fault Injection & Diagnostics Harness | *RTDE, URSim, Fault Injection*

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

Castline Studio | *Resin 3D Printing, Next.js, TypeScript, WebAssembly*

- Operate resin 3D-printing production for an e-commerce business generating \$2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

Research Publications

Do Published HumanEval Rankings Survive Local Deployment?

- Reproduced a five-model HumanEval ranking on an RTX 4070 and identified a local ranking reversal; released as a technical report.

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that acceleration depends on delegate and threading configuration; archived on Zenodo.

Conditioning and Directionality in Adversarial Transfer on CIFAR-10

- Evaluated directional transfer across three CNN families with multiple seeds and denominator-aware metrics; revising the manuscript following a TMLR rejection.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; submitted to the CoRL 2026 LfC workshop.

Smoothness Ranks Skill, Not Success

- Audited published trajectory-smoothness metrics under episode-length controls and found they primarily rank operator skill rather than task success; submitted in venue-specific forms to three workshops.

Technical Skills

Hardware & Diagnostics: Electrical and mechanical troubleshooting, wiring and harness assembly, multimeter diagnostics, actuator control, soldering and PCB rework, fault injection, root-cause analysis, telemetry analysis

Robotics & Embedded Systems: RTDE, URSim, CAN/SocketCAN, teleoperation, real-time control, LeRobot, OpenPI

Machine Learning & Scientific Computing: PyTorch, JAX, LoRA, NumPy, pandas, scikit-learn, ONNX, model evaluation

Languages: Python, C++, TypeScript, JavaScript, SQL, Bash

Systems & Data: Linux, Git, Docker, Kubernetes, Redis Streams, PostgreSQL, REST APIs, AWS

Education

Grand Canyon University	May 2024 – Oct 2025
<i>M.S. in Computer Software Engineering</i>	<i>Phoenix, AZ</i>
• Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.	

University of Silicon Valley	May 2021 – Aug 2023
<i>B.S. in Computer Science</i>	<i>San Jose, CA</i>
• Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.	

Certifications

Accelerated Computing in Modern CUDA C++ (SepHackerRank Software Engineer Certificate (Mar 2026) 2026)

Cisco Network Support and Security (Sep 2025) **IBM Artificial Intelligence Fundamentals** (Aug 2025)

Stanford AI and Machine Learning in Healthcare (JunGoogle Foundations of Project Management (Sep 2024) 2025)

Google IT Support Professional Certificate (May 2021)